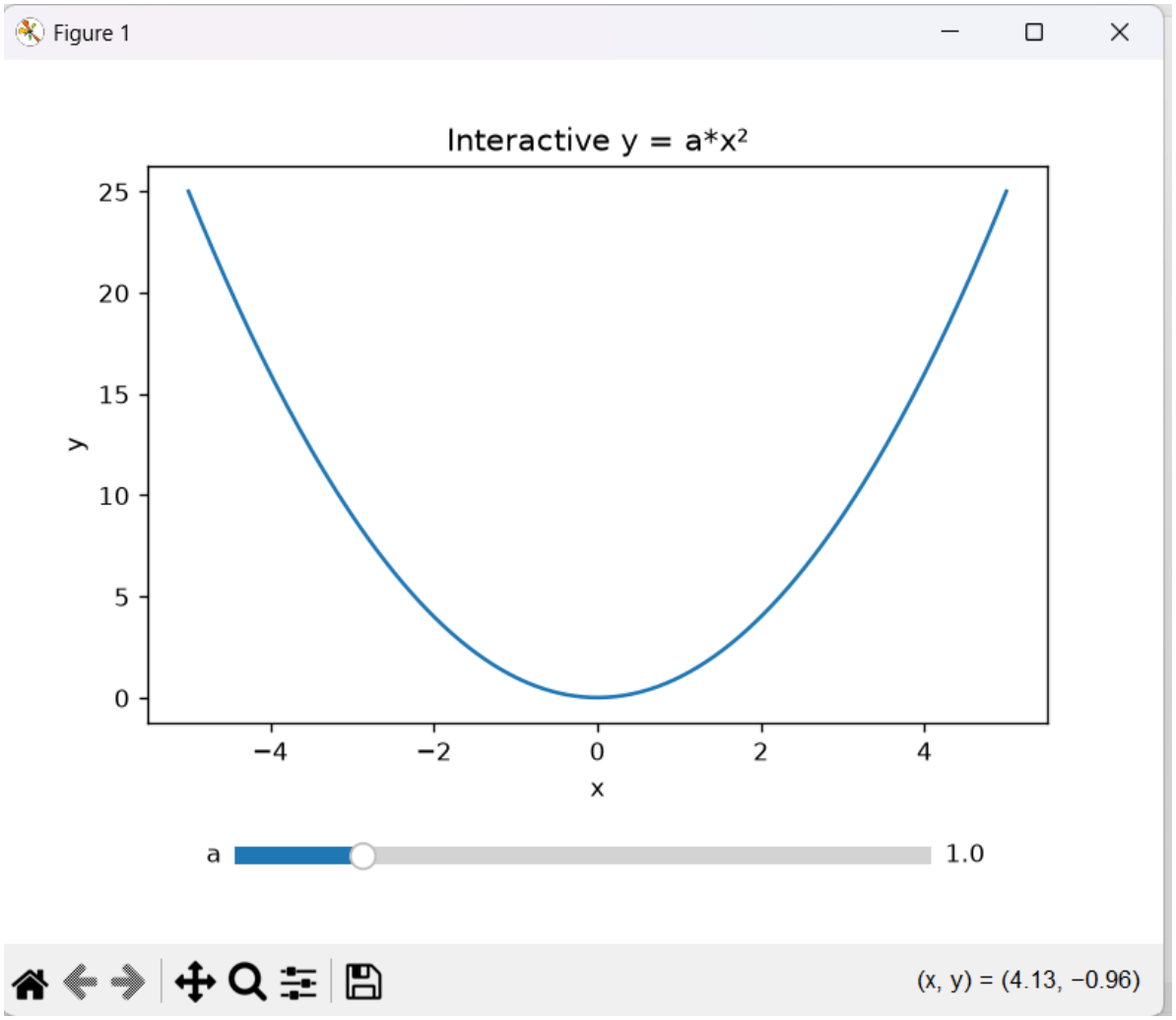
```
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.widgets import Slider
x = np.linspace(-5, 5, 200)
a = 1
fig, ax = plt.subplots()
plt.subplots_adjust(bottom=0.25)
# Initial curve
line, = ax.plot(x, a * x**2)
ax.set_xlabel("x")
ax.set_ylabel("y")
ax.set_title("Interactive y = a*x²")
# Create slider
slider_ax = fig.add_axes([0.2, 0.08, 0.6, 0.04])
slider = Slider(slider_ax, "a", 0.1,5.0,valinit=1)
# Update curve
def update(val):
  a = slider.val
  line.set_ydata(a * x**2)
  fig.canvas.draw_idle()
slider.on_changed(update)
```


Figure 1
Interactive y = a*x²
25
20
15
10
5
0
y
−4
−2
0
2
4
x
a
1.0
(x, y) = (4.13, −0.96)